

EN.553.740 Machine Learning 1

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September 21, 2026

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0.1 Introduction

A good way to think of machine learning is instead of a computer taking data and a program to produce output, it takes data and output and attempts to produce a program, ie "learn" a function.

We use it for tasks that are too complex to program, like driving or large scale data analysis. A bit of history of "learning":

- 1959: Arthur Samuel (IBM) created a program to play checkers, introducing rule-based learning.
- 1960: Frank Rosenblatt introduced a linear classifier based on brain neurons, called a Perceptron.
- 1969: Marvin Minsky showed limitations of the perceptron, halting development for 20 years
- 1980: First workshop in machine learning
- 1986: Multilayer networks and backpropagation introduced
- 1990: Vapnik and Chervonenkis introduce kernels and SVM, and termed "Statistical Learning"
- 2002: Rise of graphical models
- 2010: Deep learning takes over
- 2016: ML hits record size with NIPS and ICML doubling in size

Machine learning has been used for a variety of tasks, like stock market prediction, decoding thoughts from brain scans, and self-driving cars.

0.1.1 Learning Problems

Statistical Learning falls under five rough fields:

- Classification (Assign a category)
- Regression (predict a real value)
- Ranking (order items)
- Clustering (item partitioning, like image segmentation)
- Dimensionality Reduction/Manifold Learning (Computer vision)

In a broader sense, there is supervised learning (Classification/regression) and unsupervised learning (Clustering, dimensionality reduction, density estimation) and things like semi-supervised learning, active learning, reinforcement learning, etc.

Supervised Learning For a given problem, we learn a function $f : X \rightarrow Y$ such that $x \in \mathcal{X}$, $y \in \mathcal{Y}$. Classification deals with discrete labels while regression deals with continuous functions.

Unsupervised Learning For a given problem, we learn a function $f : X \rightarrow Y$ such that $x \in \mathcal{X}$, $f(x) \in \mathbb{R}$. Usually embedding (for text), clustering, or density estimation.

0.2 Mathematical Review

Algorithms are built from sets, functions, vectors, matrices, norms, and derivatives, so we introduce common notation here. Consider a student success dataset with the target being quiz score, and features as study time and prior score.

0.2.1 Sets and Power Sets

Let $A = \{1, 2, 3, 4, 5\}$ be the set of students in the toy dataset. The *power set* is $\mathcal{P}(A) = \{B : B \subseteq A\}$. If $|A| < \infty$, then $|\mathcal{P}(A)| = 2^{|A|}$. A subset $B \subseteq A$ may represent a mini-batch, validation set, or a selected group of samples.

0.2.2 Functions

For sets A, B , write $B^A = \{f : A \rightarrow B\}$. if $B = \mathbb{R}$, then $\mathbb{R}^A$ is the set of real-valued functions on A . If $|A| < \infty$, then $\mathbb{R}^A \simeq \mathbb{R}^{|A|}$ as vector spaces. For our toy example:

$$x_1 : A \rightarrow \mathbb{R} \quad x_2 : A \rightarrow \mathbb{R} \quad y : A \rightarrow \mathbb{R}$$

For a subset $C \subseteq A$, define an *indicator function*:

$$\mathbb{1}_C : A \rightarrow \{0, 1\}, \quad \mathbb{1}_C(i) = \begin{cases} 1, & i \in C, \\ 0, & i \notin C \end{cases}$$

Indicators are used to encode events, groups, labels, masks, etc.

0.2.3 Vectors